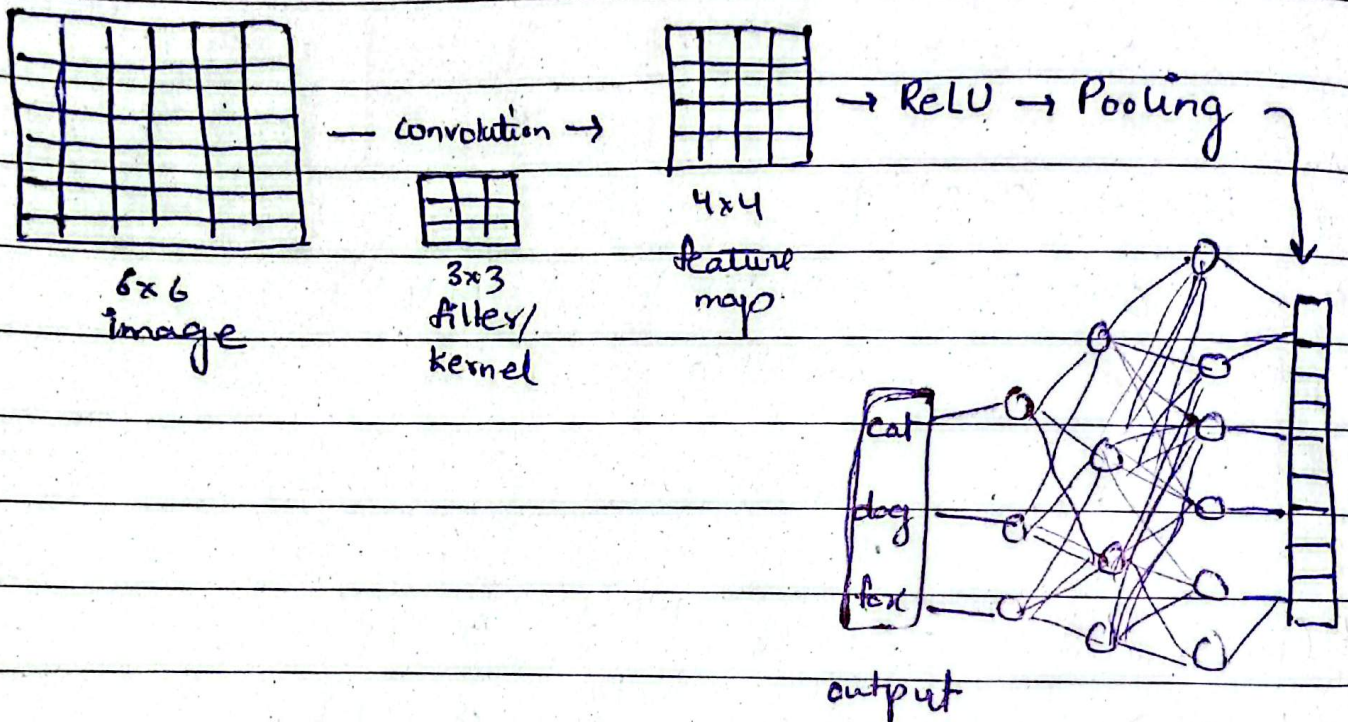


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CONVOLUTIONAL NEURAL NETWORK



Step 01: 6x6 Image

Kernel (3x3)

1	2	0	3	1	2
4	1	2	1	0	1
1	3	1	2	2	0
2	0	1	3	1	1
1	2	2	1	0	2
0	1	3	2	1	0

$K =$

1	0	-1
1	0	-1
0	0	-1

→ how many steps the kernel would move

Stride = 1 , padding = 0

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$$\text{Output} = \left\lfloor \frac{\text{Image} - \text{Kernel} + 2p + 1}{s} \right\rfloor = \frac{6 - 3 + 2 \times 0 + 1}{1} = 4$$

Step #02: Applying Convolution

$$\begin{bmatrix} 3 & 0 & 0 & 3 \\ 3 & -2 & 1 & 4 \\ 0 & -1 & 1 & 3 \\ -3 & -3 & 4 & 3 \end{bmatrix}$$

Step #03: Apply ReLU $\rightarrow \max(0, x)$ ↗ Simply -ve value
to 0 or denge

$$\begin{bmatrix} 3 & 0 & 0 & 3 \\ 3 & 0 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 4 & 3 \end{bmatrix}$$

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Step # 04 : Apply Max Pooling $\rightarrow 2 \times 2$

stride = 2

$$\begin{bmatrix} 3 & 4 \\ 0 & 4 \end{bmatrix}$$

$\hookrightarrow$ In a 2×2 region pick the max element